

Helix VPN — Session Continuation File

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Helix Constitution §11.4.131 — standing session-resumption artifact.
Re-read this file at the start of any new session before touching code.

Summary

Branch: main (single working branch). **Overall status:** Full MVP specification set COMPLETE (11 volumes V0–V10, 126 .md files with synced HTML/PDF siblings). Workable-items SQLite DB populated (484 items). Constitution fully integrated. No VPN application code exists yet — spec-only mandate.

Active work (2026-06-26):

1.  §11.4.134 re-review — manual confirmation GO (proto consistent, D8 present, spine correct)
2.  §11.4.168 Mermaid rendering — FIXED, all 126 HTML clean, 98 with embedded diagrams
3.  §11.4.93 workable-items SQLite DB — 484 items loaded
4.  §11.4.106 docs_chain wired — 2 contexts, doctor OK, sync in-sync
5.  §11.4.65 HTML/PDF exports — all docs have synced siblings
6.  §11.4.153 DOCX exports — pipeline updated, generating DOCX alongside HTML/PDF
7.  Go workable-items binary (HVPN-P1-150) — subagent building
8.  vasic-digital component repos — subagent creating on GitHub+GitLab
9.  Pre-build gate updated — Inv7 (DB) + Inv8 (docs_chain) added

10.  .gitignore-meta regen mechanisms created (§11.4.77)
11.  README Tracked-Items section added (§11.4.57)
12.  Script companion docs created with HTML/PDF siblings (§11.4.18)

Spec location: docs/research/mvp/final/ — see MASTER_INDEX.md for the full document tree.

Completed Work (highlights)

1. Constitution submodule + mandatory submodules

- constitution/ → HelixDevelopment/HelixConstitution.git (branch main)
- 11 own-org repos under submodules/<name> (flat, lowercase snake_case)
- install_upstreams run in each; .helix-manifest.yaml audit record
- Pre-commit hook, CI DISABLED (§11.4.156), local enforcement active

2. Full MVP specification set (V0–V10)

- 11 volumes, ~140 nano-detail documents, ~11.7K lines in the spine + pass-1 set
- All 16 research docs cited; decisions D1–D8 surfaced
- 46 Mermaid diagrams, SQL DDL, Podman/Docker/K8s manifests
- Every volume adversarial-reviewed (§11.4.142) + reconciled to GO (§11.4.134)
- 126 .md / 126 .html / 126 .pdf — all synced (§11.4.65)

3. Workable-items SQLite DB (§11.4.93)

- docs/workable_items.db — 484 items (P0: 36, P1: 210, P2: 132, P3: 96)
- Schema: items, item_history, test_diary, gates, operator_block_details, obsolete_details, meta
- Loader: scripts/workable_items_loader.py (md-to-db, bidirectional)
- All items start as Queued / Task status

4. Research corpus

- docs/research/mvp/ — 16 source docs (11 LLM analyses + 5 refined)
 - v09-research/ — 10 per-angle research dossiers (all cited, all verified except wireguard partially)
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What Remains

Pre-tag quality items

1. **D-PKI-CA-TIER** — two-tier issuing CA (reconciled to svc-pki source); operator may veto for single-tier MVP
2. **D-OD-1** — OpenDesign interpretation (authoring layer + decoupled token export); awaits operator confirm

In progress (subagents)

1. **Go workable-items binary** (cmd/workable-items/) — HVPN-P1-150, subagent building
2. **vasic-digital component repos** (GitHub+GitLab) — subagent creating

Deferred

1. **Implementation phase** — spec-only mandate; no code until operator directs

Evidence Locations

Artifact	Path
MVP spec set	docs/research/mvp/final/ (126 md/html/pdf)
Master index	docs/research/mvp/final/MASTER_INDEX.md
Spec spine	docs/research/mvp/final/SPECIFICATION.md
Research corpus	docs/research/mvp/ (16 source docs)
Research dossiers	docs/research/mvp/final/v09-research/
Workable-items DB	docs/workable_items.db (\$11.4.93/.95)
DB loader	scripts/workable_items_loader.py
DB loader docs	docs/scripts/workable_items_loader.md
docs_chain wrapper	scripts/docs_chain_md_to_db.sh
docs_chain contexts	.docs_chain/contexts/*.yaml
.gitignore-meta	.gitignore-meta/*.yaml (\$11.4.77 regen mechanisms)
Pre-build gate	tests/pre_build_verification.sh (8 invariants)

Artifact	Path
Export script	scripts/testing/sync_all_markdown_exports.sh
Mermaid helper	scripts/testing/render_mermaid_blocks.py
Mermaid cache	.mermaid-cache/ (content-addressed PNGs)
Constitution	constitution/ (submodule)
Submodule audit	.helix-manifest.yaml
Pre-commit hook	.githooks/pre-commit
CI (DISABLED)	.github/workflows/constitution.yml.disabled-local-only

Resumption prompt (§11.4.127)

SHORT variant

Continue work on main in /Volumes/T7/Projects/helix_vpn; read docs/CONTINUATION.md first, then check background tasks and continue the spec-materials work queue.

FULL variant

You are resuming work on the Helix VPN project.

Repository: /Volumes/T7/Projects/helix_vpn
 Branch: main
 Handoff doc: docs/CONTINUATION.md ← read this FIRST

State at handoff

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- Full MVP spec set: 126 md/html/pdf under docs/research/mvp/final/
- Workable-items DB: docs/workable_items.db (484 items, §11.4.93)
- Constitution: constitution/ submodule active
- CI: DISABLED (§11.4.156), local enforcement via .githooks/pre-commit
- No VPN application code exists (spec-only mandate)

Active work queue

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- 1. §11.4.134 re-review of V1/V6/V7/V8/V0 reconciliation fixes
- 2. §11.4.168 Mermaid rendering verification (full re-export)
- 3. docs_chain wiring (§11.4.106)
- 4. vasic-digital component repos (deferred until spec stabilizes)
- 5. DOCX exports (§11.4.153)

First actions

1. `git fetch --all`
2. Check background task status
3. Continue the work queue from item 3+